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Review of experimental procedures and modelling techniques for flow behaviour and their relation to residence time in constructed wetlands

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ABSTRACT

Accurate modelling of constructed wetlands is vital for furthering the use of these systems. Constructed wetlands can treat various types of wastewater, including domestic wastewater, agricultural wastewater, industrial and mining wastewater and stormwater runoff. Physical, chemical and biological processes all take place simultaneously to breakdown the contaminants in wastewater, and while the combination of so many processes makes them an effective wastewater treatment system, it can also make the modelling of these systems intricate. The modelling technique used is dependent on the goal of the model, whether it is for design or scientific research. The extent to which contaminants are broken down by these processes is highly dependent on the time the water spends in the system, and this time is dependent on and can be quantified by the understanding of the hydraulic behaviour of the constructed wetland. Tracer studies are the most common way to determine the hydraulic behaviour of the system and the data from these studies can be used in various ways to quantify time in the system. There are three aims of this paper, the first is to give wetland researchers an idea of what hydraulic modelling techniques are available, the second is to indicate the connection between accurate hydraulic modelling and the modelling of kinetic processes, and the third is to guide the researchers in determining which model is best suited for their needs based on research needs and characteristics of the wetland.

1. Introduction

Constructed wetlands (CWs) are low-cost and low-maintenance systems which have the ability to treat various types of wastewater including domestic wastewater [1–3], agricultural wastewater [4,5] and stormwater run-off [6], to name a few [7]. There are three main types of CWs, free water surface (FWS), horizontal sub-surface flow (HF) and vertical flow sub-surface (VF). Physical, chemical, and biological processes all take place simultaneously within the CWs and in doing so reduce the pollutant load of the wastewater. To use these systems as an effective form of wastewater treatment, accurate design and modelling are required. There are various CW design methodologies which are primarily linked to flow type. They can include characteristics such as surface or subsurface flow, vertical or horizontal flow and saturated or unsaturated flow [8]. For these systems to be effective, accurate design and modelling are required. Many studies have been conducted in the last three decades to improve the models, with varying levels of complexity. The problem with CW modelling is that the inherent complexity of the CW itself means that the model may require more time or computational power than is available or necessary under the circumstances [8]. For the design of CWs, the models developed tend to be simpler and require less computational power than those developed for scientific research, but they may not be as accurate [8].

1.1. History of wetland modelling

The original CW models developed were considered 'black-box' models which relied on empirical correlations between inlet and outlet concentrations [9] and were often developed for, and could only be used for, a very specific system. Correlations based on the surface area, regression and first-order models were used and inherent in these models is an assumption about the flow, typically that there is ideal plug flow in the CW. Given the complexity of these systems and based on research conducted in recent years, it can be said that this assumption does not hold and that these empirical correlation-based models are not adequate to model CWs. Subsequently, modelling has advanced to mechanistic and process-based models. These models can be divided into two categories:

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1 Models which describe the hydraulic behaviour of the CW in detail but assume simple kinetics or focus on the reactive transport of a single contaminant
2 Models which describe the detailed reactive transport of a single (or multiple) contaminant(s), but assumes simplified or ideal hydraulic behaviour

In both cases, some simplifying assumptions have been made, either with regards to reaction kinetics or hydraulic behaviour. In recent years, models have been developed which combine detailed modelling of both hydraulic behaviour and reaction kinetics. These models typically require extensive computational power, a specific software, and a detailed understanding of numerical modelling.

CWs are, in essence, a reactor and the extent of reaction is dependent on the time the fluid spends in the CW. An investigation into the hydraulic behaviour of a CW allows for the determination of this time and subsequently the effectiveness of the system in reducing contaminant load. Therefore, for CW models to be accurate the characterisation of hydraulic behaviour must be accurate. The flow behaviour of a CW influences its ability to breakdown inlet contaminants [10]. There is a distribution of flow velocities within the system and the combination of these gives the overall hydraulic behaviour of the system. Many investigations have found that there are 3 different hydraulic zones in CWs; the main flow path, a temporary storage zone, where components and water are exchanged with the main flow path, and a dead zone [11, 12]. By investigating the hydraulic behaviour of a CW it is possible to indicate whether there are hindrances to ideal flow such as bypassing, which is when a section of the fluid moves through the system much more quickly than the bulk of the fluid, resulting in reduced time spent in the CW, dead zones, which are sections of in the system through which fluid does not flow and solute transport is due only to dispersion and mixing. All these factors contribute to the time that wastewater spends in the CW, known as the residence time. This in turn relates to how long the water is in the system to break down organics, convert nitrogen species and degrade other contaminants. If a CW has poor hydraulic behaviour, then pollutant removal will be poor and conversion for which the system was designed, will not be achieved.

It is tempting to model the flow of a CW as an ideal Plug Flow Reactor (PFR), in which every fluid particle moves through the reactor with the same velocity and has the same residence time. However, it has been shown by multiple researchers that the hydraulic behaviour of CWs is far from this ideal case [13,14]. The flow characteristics of CWs are elaborate and non-ideal owing to the changing root structure and the complicated flow pathway of fluid through gravel or soil matrix (Bonner et al., 2017). Factors such as CW shape [15,16], inlet and outlet location [17,18] and vegetation distribution [17,19–21] can affect the hydraulic behaviour of CWs. Vegetation distribution is a variable barrier to flow in terms of the plant size, shape and position. As a result, there is bypass and short-circuiting around plant mass, forming dead zones, which leads to a hold up of water results zones with no reaction in certain areas. The formation of different velocity profiles can lead to a distribution of residence times. The shortest time is experienced by the water moving at the maximum velocity in the profile, which would normally be in the surface layer of the micro-channels, with the longest times experienced by water that moves near drag-inducing surfaces, such as the CW bottom [13].

There are various techniques and models which exists for the characterisation of the hydraulic behaviour of CWs and the solute transport in the systems. These include the method of moments technique, the tanks-in-series reactor model, the advection-dispersion transport equation etc. A review of the experimental methods used to obtain hydraulic characteristics, the methods used to analyse the data from these experiments and the use of this data in process-based models, is the focus of this paper. This information is important since the hydraulic characteristics determine the length of time the wastewater spends in the CW and the time determines the extent of treatment. A guide to conducting investigating and modelling the hydraulic characteristics is included to aid the design and effective use of CWs as a form of wastewater treatment

2. Experimental procedures for understanding flow behaviour

The most common way to characterise the flow patterns and assess the hydraulic behaviour of a CW is by conducting residence time distribution (RTD) studies [22]. The residence time distribution of a reactor gives the probability of a fluid element spending a specific amount of time in the reactor and is characteristic of the mixing that occurs within the system [23]. The RTD is obtained by running a flow tracer study on the system. A flow tracer study is a stimulus-response experiment in which an inert flow tracer is injected either as an impulse or as a step-change in concentration into the system inlet pipe and the tracer concentration is measured continuously at the effluent. An effective tracer study requires the flow to be laminar [24] if the data is to be successfully extrapolated (scaled) to other operational flow rates (as long as they are also laminar). Data from these experiments are then used in RTD modelling methodologies to analyse the flow characteristics of the reactor and can be used to quantify a set of characteristics that describe the hydraulic behaviour of CWs [25]. In order for these tracer tests to be effective the tracer used must be conservative, i.e. should not be consumed or degraded [26]. The tracer should also not be susceptible to biological, chemical or photochemical decay, should be highly soluble in water, and should be easily detectable and not exist at high background levels within the system. It is also assumed that for any experiment run that the tracer flow behaviour is similar to a normal effluent liquid [27]. The main different tracer techniques are discussed here.

2.1. Impulse-response tracer tests

The type of tracer study most commonly used to determine the hydrodynamic behaviour of a CW is the impulse-response test [22,28]. For conducting an impulse-response test a small volume of highly concentrated tracer solution is injected instantaneously (equivalent to a mathematical impulse function) into the inlet of the system and the concentration of the tracer is measured at the effluent at regular time intervals [23]. In Fig. 1 a graphical representation of the concentration-time response at the inlet (a) and outlet (b) can be seen. The very short impulse is observed to spread (or disperse) through any system. If the resulting distribution is normal (Gaussian), the flow is normally ideal. The shape of the distribution can tell the experimenter many things: if it is bi- or multi-modal (two or more peaks) it can indicate bypass, if it has a very long tail, it can indicate dead-zones as the tracer 'bleeds' back from the dead-zone into the outlet.

The residence time distribution function, E(t), which is a quantitative description of how much time the fluid particles have spent in the reactor, can be obtained from concentration-time data of the impulse-response curve and is given by Eq. 1. The assumption for Eq. 1 is that the inlet volumetric flow rate of the system is constant. E(t) is calculated using the area under the concentration-time curve which, to normalize the distribution, the area under the E(t)-time curve is unity. The calculation of E(t) is important as the first step towards calculating the moments of the distribution (i.e. the first moment is the mean of the RTD, the second moment is the spread of the distribution – these are discussed further later in this paper).

$$E(t) = \frac{C(t)}{\int_0^\infty C(t)dt} \tag{1}$$

Two of the problems associated with an impulse-response tracer study is ensuring that the injection of tracer takes places over a period of time which is short compared to the residence time of the system, and ensuring negligible dispersion of the tracer between the point of injection and the feed to the system. Early truncation of an impulse-response

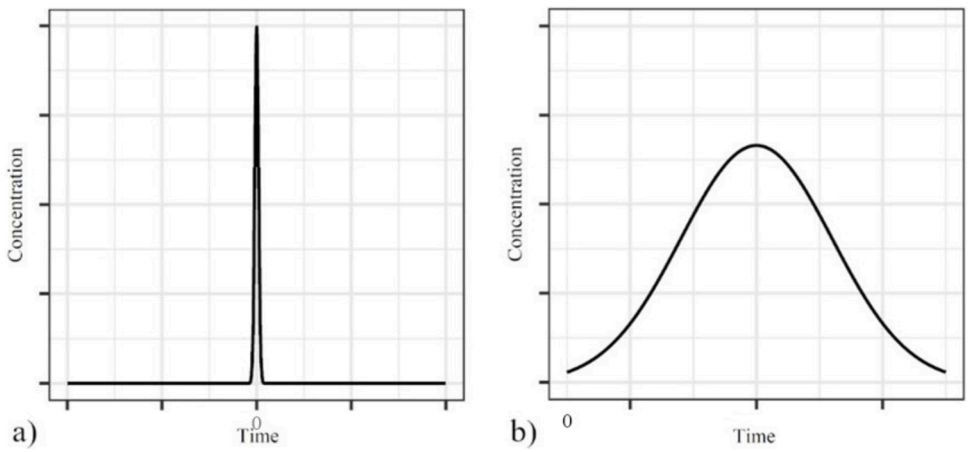

Fig. 1. A graphical representation of the concentration-time response of an impulse-response tracer test: inlet (**a**) and outlet (**b**).

tracer study can lead to inaccuracies as this method is sensitive to the concentration-time data at the tail of the response curve. When conducting these experiments, it is important to ensure this doesn't happen. However, if early truncation does occur then the researcher can extrapolate the tail of the response curve. A further major obstacle with the impulse-response tracer test is that the same concentration may occur at various times (i.e. on the way towards the peak(s) and on the way down from the peak(s). This means that the actual time a molecule has spent in the reactor is not known, and rather there are probabilities of time that reactants spend in the reactor.

2.2. Step change tracer tests

The other primary type of tracer study is a step-change response test. For this experiment, a constant rate of tracer is fed into the system starting at $t = 0$. The concentration of this tracer must be significantly higher than the background concentration and easy to identify analytically. The effluent concentration is monitored at regular time intervals [29] until the outlet concentration is stable and equal to that of the inlet (this will only happen if the tracer is conservative – if not the outlet will reach a steady-state value). A graphical representation of the concentration-time behaviour of the typical inlet and outlet response of step-change tracer tests is shown in Fig. 2.

From the concentration-time data of the step-change response experiment, the cumulative distribution, F(t), can be obtained. F(t) is described as the fraction of effluent that has been in the system for less than time t. The mathematical definition of the cumulation distribution is given in Eq. 2 [23]. F(t) can also be considered the normalised response to the step input.

$$F(t) = \left[\frac{C_{out}}{C_{inlet}}\right]_{step} \tag{2}$$

The F(t) and E(t) curves are mathematically related. Similarly, the cumulative distribution function is necessary to calculate parameters such as mean residence time and dispersion.

Some of the challenges associated with step-change response tracer studies include difficulty maintaining a constant feed concentration, a large amount of tracer is required to do this, and the analysis of step-change response data involves differential mathematics rather than integral mathematics. This can result in lead to large errors and a serious increase in noise in the data the slope of the response curve is being measured rather than the area underneath it [23]. By measuring the slope, the data is much more sensitive to analytical uncertainty. One of the benefits of this type of study, however, is that it is not necessary to know the total amount of tracer added to the system as it tends towards a known steady-state outlet.

The relationship between the residence time distribution function (E(t)) and the cumulative distribution function, is given in Eq. 3. This integral relationship allows for the conversion from impulse-response data to step-change response data if desired.

$$\int_0^t E(t)dt = F(t) \tag{3}$$

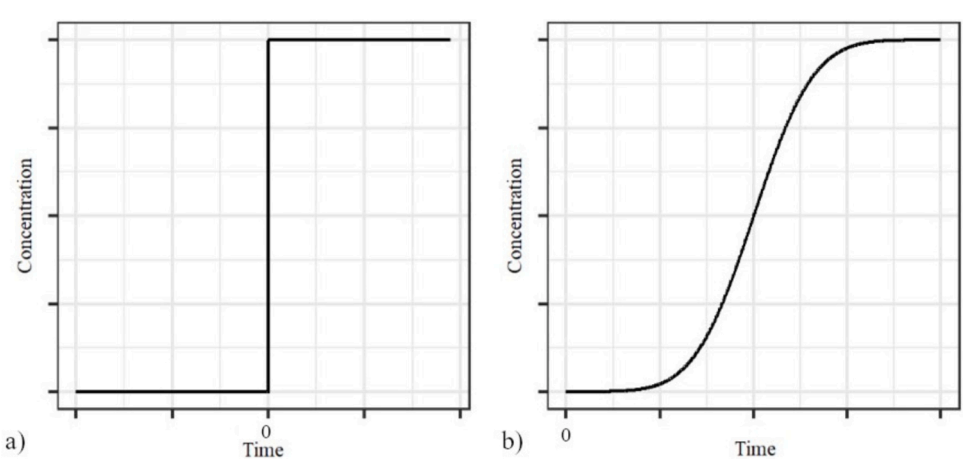

Fig. 2. A graphical representation of the concentration-time response of a step-change response tracer test: inlet (**a**) and outlet (**b**).

2.3. Square input tracer tests

Square input tracer tests are uncommon and not well covered in the literature. In this form of tracer study, the feed concentration is increased from 0 to a significantly higher and measurable concentration at t = 0 and then at a t = t* the feed concentration is stepped back down to 0. Giraldi et al. (2009) used a square input tracer test when conducting tracer tests on VF CWs to determine the hydrodynamics in different saturation conditions [30]. The square input signal was artificially extended to take the form of a step change input signal using Eq. 4, where C(t) is the normalized tracer concentration and t* is the time and where the dosing of tracer for the square input was stopped. In other words, it is essentially a double step-change – one up, and one down.

$$\begin{cases} F(t) = C(t) & , t \leq t^* \\ F(t) = C(t) + F(t - t^*) & , t > t^* \end{cases} \quad (4)$$

2.4. Tracer options

There are many options of tracers available which include but are not limited to salt ions, fluorescent dyes and stable isotopes, or a combination of these of tracers. All of these options have been used by researchers conducting tracer tests in CWs.

2.4.1. Salt tracers

Salt tracers, also known as ionic tracers, are considered to be effective for studying the hydraulic behaviour of CWs as they are easily soluble and are unlikely to interact with the biota of the system [31]. Sodium chloride is one of the most common salt tracers and has been used by many researchers. Heiderscheidt et al. (2020) used it to determine the residence time under frozen and frost-free conditions of an unplanted, peat-based, pilot-scale CW [32], Sun et al. (2019) used it to determine hydraulic parameters of an HF [33] and Hua et al. made use of the tracer to investigate the effect of clogging and resting processes on the flow behaviour of a VF CW [34]. In all three studies, the researchers measured the electrical conductivity (EC) at the outlet of the CW and based on a linear correlation between EC and the concentration of sodium chloride, determined the concentration of the tracer at the outlet of the system. One study used lithium as a tracer to validate the hydraulic model used in CWM1-RETRASO mechanistic model in the simulation of an HF CW [2], while another used lithium chloride to test the effect of inlet configuration and vegetation type on the hydraulic performance of 18 (FWS) CWs [35]. In the latter study atomic spectrophotometry was used to analyse the concentration of Li^+ ions after HNO3 was added for sample preservation and KCl was added as an ionization suppressant. Potassium iodide was used by a group of researchers to determine the hydraulic behaviour of drained peatlands [36]. Analysis of the tracer test samples was done using an iodine selective electrode to determine the concentration of iodide [36].

One of the limitations of salt tracers is that the salt solutions typically have higher densities than the CW water [35] and so a halocline could be formed. This results in the stratification of fluid which prevents mixing and is thermodynamically metastable. Density induced stratification may be enhanced by low flow velocities [37]. Special care needs to be taken, especially in the case of impulse-response tracer tests, to ensure that salt tracer settling does not take place. Density differences are also affected by temperature, but in FWS CWs the temperature difference from top to bottom is approximately 1 °C which is not considered enough to create stratification [10]. Any temperatures differences greater than this may cause a problem [35].

When choosing a salt tracer, choosing one with ions that have minimal interaction with the system is important. For example, Cl^- and Li^+ are good ionic tracers but NO_3^- will be degraded in the system and therefore is not a conservative tracer. It is also important to consider the charge of the ion and the charge of the CW medium, to ensure minimal interaction between the two.

Analytical methods most commonly used for salt tracers are electrical conductivity and ion chromatography and, in some cases, atomic absorption spectrophotometry. Electrical conductivity is a cheap and effective method of analysis and can be used as a proxy for rapidly determining the concentration of ions in the system.

2.4.2. Fluorescent tracers

The use of fluorescent tracers in the RTD studies of CWs is common. Fluorescent dyes such as fluorescein and rhodamine are inexpensive, easy to handle and easy to detect [38]. One study used rhodamine WT to evaluate hydraulic behaviour and performance indicators of various CWs designs and from this information the researchers were able to choose the optimum design from the systems tested on [39]. An inline optical fluorometer YSI 6130 RWT sensor installed on a YSI multi-parameter water quality sonde was used to determine the concentration of tracer at the outlet. Guo et al. (2019) used rhodamine WT to determine the residence time and other hydraulic parameters of a tundra CW and from this were able to determine first-order rate constants of the contaminants which can be used to further improve the CW design [40]. Another common fluorescent dye tracer is uranine. Pálfy et al. (2017) used uranine to determine the hydraulic behaviour of a CW treating combined sewer overflow and from this information the researchers were able to identify short-circuiting as a limitation of the system [41]. In this study, an inline fluorometer (GGUN-FL932) was calibrated and used to determine and record the concentration of tracer at the outlet. Laurent et al. (2015) used uranine and sulforhodamine B to successfully determine the hydrodynamic behaviour of three FWS

CWs [42]. An inline fluorometer (GGUN-FL30) was calibrated for both tracers and used to determine and record outlet concentrations. These dyes are for the most part conservative but have the potential to adsorb onto certain materials, are susceptible to photodegradation and biodegradation and can be pH sensitive [31,43].

2.4.3. Combinations of different types of tracers

A combination of two or more types of tracers is being trialled by CW researchers for RTD studies. Different tracers have different diffusion coefficients, and this allows researchers to determine the presence and location of stagnant zones and diffusion dominated transport processes [44]. Birkigt et al. (2018) conducted a multi-tracer RTD experiment using bromide, uranine and deuterium oxide to determine the flow behaviour in three dimensions of a pilot-scale HF CW [45]. The method of analysis for each of these tracers was different, ion chromatography was used to determine the concentration of bromide, a fluorescence fluorometer was used to determine the concentration of uranine and high-temperature pyrolysis was used to determine the concentration of deuterium [45]. From this research, it was noted that bromide and deuterium oxide had similar behaviour and the data from the tracer studies yielded similar hydraulic parameters while uranine displayed different hydraulic behaviour. It was suggested that there was a lag in the response of uranine was due to organic matter in the system. Mailard et al. (2016) used bromide, uranine and sulforhodamine-B as tracers to determine the impact of batch versus continuous-flow modes on the dissipation of the chiral herbicide S-metolachlor in a CW [46]. These researchers found that plant uptake, sorption and photodegradation are greater in batch mode operation that continuous flow operation, but the extent for each tracer was slightly different. Bromide was most susceptible to plant uptake, uranine was most susceptible to photodegradation and sulforhodamine-B was most susceptible to sorption [46]. In this case, the methods of analysis were ion chromatography for bromide and fluorescence spectrometry for both the sulforhodamine-B and uranine. Pugliese et al. (2020) used uranine and potassium bromide as conservative tracers to study the hydraulic behaviour of a shallow-deep compartment-designed FWS CW and were able to identify the presence of multiple flow paths and internal mixing in shallow zones [47]. Montalván et al. (2017) used natural tracers (Cl^-, Br^-, Na^+, Mg^{2+}) and environmental isotopes (^{18}O, ^{2}H, ^{14}C, ^{13}C and ^{3}H) to estimate residence

times of brine waters and identify recharge areas of the different flow subsystems in a hypersaline CW [48]. The environmental isotopes were analysed by isotope ratio mass spectrometry and the natural tracers were analysed using PTE-AG [48]. Bruun et al. used bromide and tritium (in the form of ^3H_2O) to investigate intra-granular diffusion in a woodchip based HF CW [49]. This study indicated that there may be a slight anion exclusion on bromine in the wood chip matrix. Ion chromatography was used to analyse the concentration of bromide ions and liquid scintillation counting was used to analyse the concentration for tritium [49].

2.4.4. Other tracer options

While the tracer types mentioned above are the more common, a few others have been used. Bonner et al. (2017) developed a mapping methodology for the use of heat as a conservative tracer in HF CWs. The researchers were able to accurately quantify the hydraulic parameters of the system being tested on from the mapped heat tracer response. It was shown that if certain parameters of transport can be determined then this technique can be used. One different study used stable isotope ratios of oxygen ($\delta^{18}O$) and hydrogen (δ^2H) in water to study the hydrology of forested CWs [50]. Through this, the researchers were able to identify the contribution of surface water and rainwater to three different CW systems and the extent of surface evaporation. Liquid Water Isotope Analyzer was used to determine the concentration of $\delta^{18}O$ andδ^2H. Abu-Bakar et al. (2017) investigated the use of microbial tracers and in this study the researchers calibrated and validated the transport and decay of four microbial tracers and used the data to refine a model for better water quality processes in a macro-tidal coastal basin. Modelling the decay of the microbial tracers proved to be more computationally intensive than the researchers expected so the technique is not considered useful for design [51]

3. Techniques used for interpretation and analysis of the residence time distribution

In an ideal plug-flow system, all fluid atoms/elements move through the CW with the same velocity and reach the system outlet at the same time. This exit time is the simplest way to account for time in a CW and is called the nominal residence time (τ) which is defined in Equation 5 [22]. The nominal residence time is only a function of volumetric flow and CW volume and is an intrinsic parameter of any reactor. This method does not account for any non-idealities in the flow through the CW system and a tracer test is not required to determine the nominal residence time. Owing to the non-ideal nature of CW hydraulic behaviour, the nominal residence time is not considered sufficient to characterise CW flow behaviour.

$$\tau = \frac{V}{Q} \tag{5}$$

3.1. Method of moments

In the method of moments technique for interpreting RTD data, three parameters are determined; the mass recovery, the mean residence time and the variance. The mass recovery of tracer in an RTD study is important, especially in an impulse-response study, as many parameters used to quantify the hydraulic behaviour of a system are determined based on the assumption of total mass recovery. Mass recovery of the tracer of 80 % or higher is considered acceptable [22]. The percentage of mass recovery of conservative tracer in a tracer study is considered the zeroth moment of the RTD. The mass recovery can be calculated using Equation 6 [52].

$$M_{out} = \int_0^\infty Q_{out}(t)C_{out}(t)dt \tag{6}$$

The first moment of the RTD study is the centroid of the response curve and is called the mean residence time (t_m), which is the average amount of time that the fluid particles have spent in the reactor. This can be calculated using Eq. 7 for impulse-response tracer studies and Eq. 8 for step-change response tracer studies.

$$t_m = \frac{\int_0^\infty tE(t)dt}{\int_0^\infty E(t)dt} \tag{7}$$

$$t_m = \int_0^\infty [1 - F(t)]dt \tag{8}$$

The second moment of the RTD is the variance. The variance is an indication of the spread of the impulse as it passes through the reactor. Eq. 9 and Eq. 10 give the equation used to determine the variance from an impulse-response and step-change tracer study, respectively. The variance indicates the global mixing in the system [30].

$$\sigma^2 = \int_0^\infty (t - t_m)^2 E(t)dt \tag{9}$$

$$\sigma^2 = 2\int_0^\infty t[1 - F(t)]dt - {t_m}^2 \tag{10}$$

Dimensionless variance is calculated using Eq. 11. The dimensionless variance indicates local mixing.

$${\sigma^2}_0 = \frac{\sigma^2}{{t_m}^2} \tag{11}$$

Another factor that is often considered but is not part of the method of moments is the effective volume utilisation. The mean residence time (t_m) can be compared to the nominal residence time (τ) to provide the effective volume utilisation (*e*). This is given in Equation 12 [35]. The effective volume utilisation is a function of aspect ratio [53], positioning of inlet and outlet ports [24], plant position and root density [54] as well as the extent of clogging in the system due to biofilm growth [55].

$$e = \frac{t_m}{\tau} \tag{12}$$

One of the limitations of the method of moments is that the anomalies in the tail of the impulse-response tracer test can lead to incorrect estimation of parameters and curve fitting [10]. Early truncation of the tail can also lead to an underestimation of the mean residence time [56]. This is especially important for packed bed reactors (such as CWs), where even those with close to ideal flow have a gaussian distribution with a long tail [57]. The long tail can lead to substantial error in the calculation of the mean of the RTD.

The method-of-moments technique can be used for FWS, HF and VF CWs as no assumption about the direction or the saturation of the flow has been made. The methods-of-moments technique was used in a study by Holland et al. (2004), to determine the sensitivity of a normalised RTD to varying flow rate and depths [52]. The researchers found that while the normalised RTDs were the same for different flow rates, the water depth influenced the RTD [52]. One group of researchers investigated the influence of the method of RTD analysis on pollutant removal estimations and hydraulic parameters by comparing the method of moments with other analysis techniques. These researchers found that the method of analysis affected the calculation of hydraulic parameters related to mixing more than other hydraulic parameters and removal predictions. In the same study, the effect of RTD truncation on hydraulic parameters was also investigated and in this section of the study the researchers noted that the truncation of tracer experiments at 3τ resulted in the overestimation of the effective volume utilisation. Another group used the method of moments to investigate the difference in hydraulic parameters of CWs constructed on previously drained peatlands compared to pristine CWs. These researchers found that in the case of the CWs built on previously drained peatlands the full volume was not being used effectively [36].

It has been indicated that a disadvantage of using the method of

moments technique is that a single mean residence time (t_m) can describe an infinite number of distributions, thereby stating that there is a 'loss of information' when determining other parameters, such as dispersion, from the mean residence time [58]. The methods-of-moments technique is also sensitive to non-continuous data sets [35]. The benefit of the method-of-moments technique is that the values of the hydraulic parameters can be obtained directly from the measured data [35]. The method-of-moments technique has been used to investigate the hydraulic behaviour of both large-scale CWs and pilot-scale CWs [36,59,60]. As this technique is based directly on data from tracer studies and no assumptions are made about the system, it can be used for CWs of any size (as long as enough tracer has been added.

3.2. Tanks-in-series model

The tanks-in-series model is a reactor model and is a gamma distribution of residence times. The model is given in Equation 13 [10]. In this model, the results from the tracer test, either step-change or impulse-response, are analysed to determine how many continuously stirred equally sized tank reactors in series would provide the same response curve as the experiment CW curve. The number of tanks in series, N, is determined using Equation 14 [61]. When N = 1 there is a large degree of back mixing and the CW is considered to behave as a CSTR and can be modelled as such. As N tends to infinity the degree of back mixing tends to zero and the behaviour of the CW tends to that of an ideal plug flow reactor (PFR) and can be modelled as such. Ideal plug flow conditions are considered optimal for wastewater treatment as the total water volume of the CWs is considered to be treated. The particles all flow through the system at the same velocity and therefore reach the outlet in the same amount of time [61]. Based on reported literature values Kadlec & Wallace (2009) estimated that N = 4.1 ± 0.4 for FWSCWs, and N = 11.0 ± 1.2 for HF CWs [10].

$$E(t) = \frac{t^{N-1}}{(N-1)!\tau_i{}^n} exp\left(-\frac{t}{\tau_i}\right) \tag{13}$$

$$N = \frac{t_m{}^2}{\sigma^2} \tag{14}$$

From the number of tanks in series another factor can be determined, the hydraulic efficiency, which is defined in Eq. 15. Hydraulic efficiency is a measure that can be used to compare different CW designs [62]. It considers the effective volume utilisation (e) as well as the degree of mixing by making use of N.

$$\lambda = e\left(1 - \frac{1}{N}\right) \tag{15}$$

One study used the tanks-in-series model to investigate the hydraulic behaviour of three large scale FWS CWs and it was determined that the vegetation density had a large effect on the dispersion of the system [42]. Another study used the tanks-in-series model with an exchange between active and dead zones to simulate the RTD of an HF CW [63]. From this model, the researchers were able to determine the optimum inlet flow rate and water level in the CW being studied. The tanks-in-series model can be used for FWS and HF CWs as in these systems the flow is saturated. The variably saturated flow which is common in VF CWs makes the use of this model more challenging, but under saturated conditions, the model can be used. It is also applicable to CWs of various sizes. Laurent et al. (2015) used the tanks-in-series model on 3 systems with areas ranging from 63 m^2 to 750 m^2 [42] while Persson &Wittgren (2003) used this model on a pilot-scale CW [63].

3.3. Advective-dispersive transport equation

The advective-dispersive transport equation has been used by many researchers to describe solute transport and flow in CWs. Unlike the tanks-in-series model, which is a reactor model, the advective-dispersive transport is a flow-based equation. In this methodology, the physical processes of advection, which refers to the transport of solute caused by the bulk flow of fluid, and dispersion, which refers to the transport of solute particles from an area of higher concentration to an area of lower concentration, are included within the transport model [64]. Advection is usually the dominant process for solute transport in groundwater [65]. The advective-dispersive transport equation is valid for solute transport in systems with impermeable boundaries after the initial mixing period, in which the solute is fully mixed with the inlet fluid. The model assumes that the solute is fully dissolved and therefore has the same average velocity as the fluid [66]. The advective-dispersive transport model with retardation and decay quantifies the hydrodynamic dispersion of a single soluble species which undergoes both decay and retardation. It is assumed that the porous media is homogeneous, isotropic and that the medium is saturated, the flow is steady-state and Darcy's Law applies [67]. Darcy's Law is a model for fluid flow through porous media which states that's linear momentum is conserved and that there is a directly proportional relationship between inlet flow rate, fluid viscosity and pressure drop over a given distance [68]. This partial differential equation can be used to describe the movement of a chemical tracer in subsurface media and is given in Eq. 16. μ and γ are the first-order decay coefficient and zero-order production constant, respectively and can be used for non-conservative reactive chemical tracers [69]. R, a chemical retardation factor, is included in the transport equation and is provided in Eq. 17. This retardation factor includes a distribution coefficient, K_d, which indicates the ratio of solute concentration in the solid and the liquid [70]. A K_d value close to 0 is considered non-sorbing and an increase in the K_d value indicates a larger sorbing tendency. The value depends on the physical characteristics of the tracer and in some cases the sub-surface media. D is the hydrodynamic dispersion coefficient and is a measure of describing the mixing of solutes in porous media [71]

$$R\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} - u\frac{\partial C}{\partial t} - \mu C + \gamma \tag{16}$$

$$R = 1 + \frac{\rho_b K_d}{\varepsilon} \tag{17}$$

The advective-dispersive transport equation is based on assumptions of saturated flow and porous media. This means that the use of this equation is limited to systems with sub-surface media and saturated flow, which in the case of CWs is most commonly HF CWs. However, the size of the CW is not a limiting factor for this model. It has been used on both large-scale CWs and pilot-scale CWs [72–74].

One study used an analytically solved form of the advective-dispersive transport equation developed by Ogata & Banks (1961) [75] to determine changing hydraulic parameters due to biological growth of two HF CWs with different filter mediums [72]. The study found that biological growth decreased the drainable porosity but had a negligible effect on hydraulic conductivity and that selecting a coarse filter medium could mitigate clogging caused by biological growth [72]. Another study used the advection-dispersion transport equation for simulation-based design of a CW to determine the optimal size under specific conditions [12]. From the simulation, the researchers determined that quantifying ambient temperature, effluent release and recycle streams were more important than precipitation and evaporation when determining the optimum size of a CW. A third study used a two-dimensional depth-averaged hydrodynamic mode combined with the advective-dispersive transport model on an HF CW to account for the effect of vegetation on the hydrodynamics in CWs [73]. The model in this study was able to predict the bimodality of the RTD response when this hydraulic behaviour was due to two different flow channels; the main flow channel and vegetated zones. Moutsopoulos et al. (2011) used the advective-dispersive transport equation to simulate the hydraulic performance and nitrogen transformation of HF CW [74]. From this model, the researchers were able to simulate nitrogen transport and decay accurately in unplanted systems but found that the extra

complexity introduced by vegetation in a planted system, either because of the root zone or uptake of ammonia by the roots, proved to be less accurate and required further investigation.

3.4. Richard's equation

Richard's equation is a partial differential equation which is used to describe unsaturated flow in a non-porous medium and is defined according to Eq. 18 [30]. The relationship between pressure head (h), hydraulic conductivity (K) and water content (θ) are described according to van Genuchten-Mualem functions (Eq. 19 and Eq. 20) [76]. This equation is commonly used to describe the water flow in VF CWs as they are often intermittently loaded and therefore the flow is variable saturated and dynamic [77]. Richard's equation forms one of the partial differential equations in the HYDRUS Wetland model, executed in HYDRUS software, which is one of the most detailed process-based models currently available [77]. Many researchers have made use of this model to successfully validate and investigate the processes taking place in CWs [78,79].

$$\frac{\partial\theta}{\partial t} = +\frac{\partial}{\partial z}\left[K\left(\frac{\partial h}{\partial z} - 1\right)\right] \tag{18}$$

$$\begin{cases} \theta = \theta_r + \dfrac{\theta_s - \theta_r}{\left(1 + |\alpha h|^{1/(1-m)}\right)^m}, & h < 0 \\ \theta = \theta_s, & h \geq 0 \end{cases} \tag{19}$$

$$\begin{cases} K = K_s\left(\dfrac{\theta - \theta_r}{\theta_s - \theta_r}\right)^{0.5} * \left[1 - \left(1 - \left(\dfrac{\theta - \theta_r}{\theta_s - \theta_r}\right)^{1/m}\right)^m\right]^2, & h < 0 \\ K = K_s, & h \geq 0 \end{cases} \tag{20}$$

3.5. VART model

Variable residence time (VART) model was initially developed as a model for longitudinal dispersion and solute transport in streams [80]. It is based on a 'double-layer' conceptual model. The upper layer is dominated by advection-transient storage and the lower layer is a diffusion-dominated storage zone [80]. The model is used for FWS CWs because they have the same 'double layer' that is conceptualised for the VART model. Additionally, streams and FWS CWs have similar transient storage zones [81]. The VART model has been used to compute the hydraulic residence time of CWs filled with two different mediums and was able to accurately predict the RTD for design purposes [11]. The model was combined with kinetic processes to simulate the removal of total phosphorus and ammonium [82], and BOD [83] in FWS CWs and it was found that there was a good simulation of the three contaminants and the combined model is considered an efficient tool for designing CWs and understanding the effects of various processes and mechanisms on the treatment of wastewater in CWs [82,83].

3.6. Convolution integral

The convolution integral can be used to describe the flow behaviour of a system. This model is not based on physical flow processes but rather the experimental data obtained directly from tracer studies. For this reason, the convolution integral can be used to describe the flow behaviour of FWS, HF and VF CWs. Sheridan et al. (2014) used the convolution integral to determine the rate constants for the degradation of ethanol and a lumped parameter, COD, in winery effluent within an HF CW [84]. These researchers found that because the rate constants obtained from the convolution integral are based directly on the RTD data, they were more accurate than the rate constants determine using the other models investigated. It is also possible to determine the conversion of a reactor, regardless of the degree of non-ideal behaviour, directly from the tracer information for first-order reactions. The convolution integral, derived by Levenspiel (1999), is given in Eq. 21 [57].

$$C_{out}(t) = \int_0^t C_{in}(t - t')E(t')dt' \tag{21}$$

It can be shown that Eq. 22 is an equivalent form of Eq. 21.

$$C_{out}(t) = \int_0^t C_{in}(t')E(t - t')dt' \tag{22}$$

The concise form is given in Eq. 23.

$$C_{out}(t) = C_{in} * E \tag{23}$$

The usefulness of this method arises from the fact that no reactor model is assumed. Rather, the unique data arising from the tracer test are directly utilised. This also means that the convolution integral can be used for CWs of any size. This method was found to be significantly more accurate at modelling the reactor processes than the PFR or tanks-in-series models.

4. Discussion

All the methods discussed in this paper have their advantages and disadvantages. Some are simple to use but do not provide detailed information about the interaction of hydraulic processes and biokinetic processes, while others allow the researcher to investigate these interactions but at a high computational cost. The choice of hydraulic depends on the need of the researcher. Fig. 3 is a tool that researchers can use to aid the decision of which model to use. If the researcher is looking to investigate the effect of changing design parameters of a CW on the mixing and the volume utilisation of a system, then the method-of-moments or tanks-in-series model would be good choices. If modelling the degradation of contaminants in wastewater and understanding the effect of hydraulic conditions on this degradation is the focus, researchers could choose from the advective-dispersive transport model, VART model or Richards equation, as these are all typically combined with biokinetic models. The type of CW being investigated will also influence the model used, if the flow is unsaturated, as is typical of a VF CW, then Richard's equation should be the hydraulic model of choice, if the system being investigated is a FWS CW then VART is the most appropriate of the models. Method of moments, tanks-in-series model, and the convolution integral can be used for all types of CWs, as they make no assumptions of the physical flow processes. The VART model, Richard's equation and the advective-dispersive transport equation do not require a tracer test, although a tracer test to validate the results from these models is recommended. This means that these models can be effective for design purposes before the system has been built.

5. Conclusion

The modelling of CWs had advanced a great deal in the last two decades, and while there has been a larger focus of the biokinetic models, the use of an effective hydraulic model is necessary for furthering research, design and use of these systems. CWs are an effective form of wastewater treatment owing to the multiple processes taking place in parallel, but this also makes them complex systems to model. The modelling technique chosen is dependent on the aim of the researcher, whether it is for understanding the effects of changing design parameters on hydraulic conditions and contaminant removal, or if it is for research purposes and investigating the relationships between multiple processes. One cannot avoid making use of a hydraulic model for either of these aims. The effectiveness of a CW in treating wastewater is dependent on the time the water spends in the system, and this time is dependent on the hydraulic behaviour of the CW. It is for this vital reason that understanding and quantifying the hydraulic behaviour of

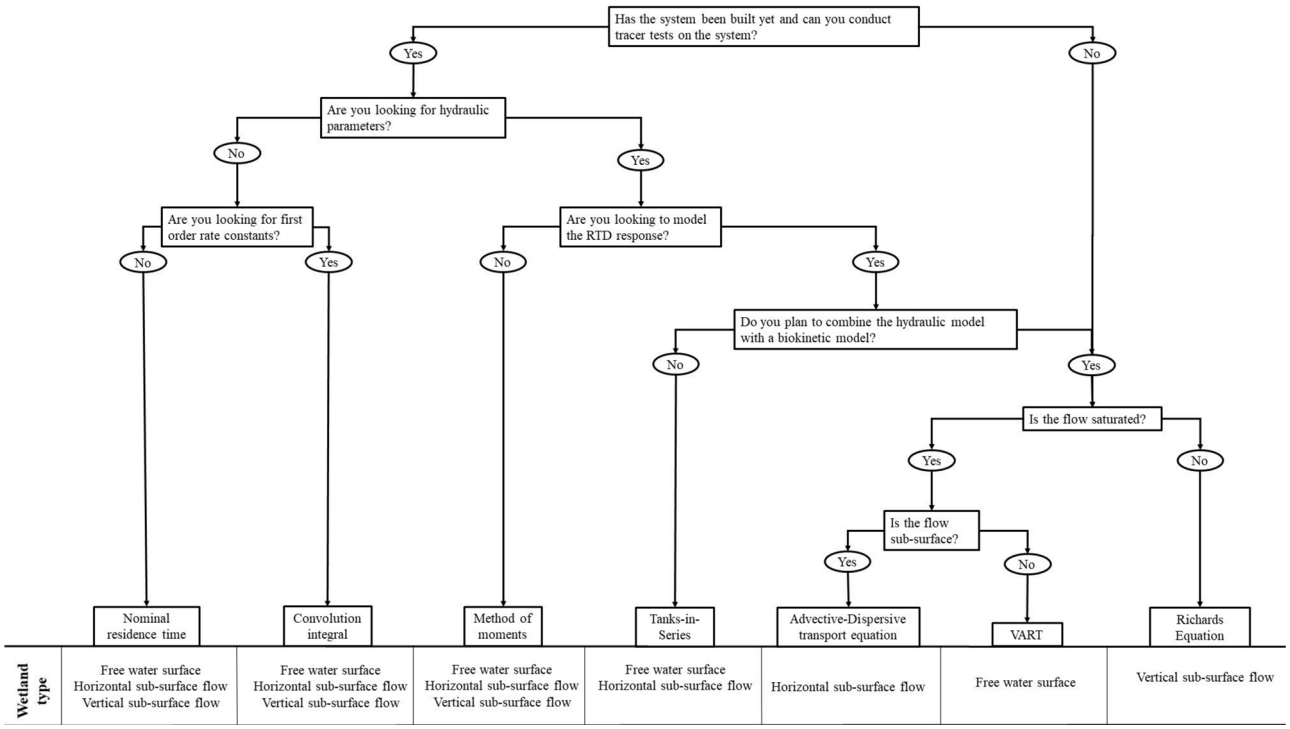

Fig. 3. Decision-making tree for choosing the hydraulic modelling technique based on research and design requirements.

CWs be achieved with the use of accurate models.

Declaration of Competing Interest

The authors of 'Review of experimental procedures and modelling techniques for flow behaviour and their relation to residence time in constructed wetlands'', Ruth Stephenson and Craig Sheridan and certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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